



The Libyan Academic Roadmap

Preparing for higher education in Libya requires careful planning. Throughout the year, an ambitious student is preparing to apply for several scholarships offered by various Libyan universities. There are N scholarships available in total.

Each scholarship i is open for applications during a specific continuous time interval $[l_i, r_i]$ (inclusive) and has an associated importance rating w_i .

The student wants to choose a single continuous period of time, denoted by the segment $[L, R]$, to work exclusively on these applications. To ensure they can make progress on every selected application, every chosen scholarship must be open for at least one day during this period. Formally, for every chosen scholarship i , the chosen period $[L, R]$ must intersect the scholarship's open interval ($[L, R] \cap [l_i, r_i] \neq \emptyset$).

To make their schedule a bit more flexible, the student is allowed to drop (skip) at most one scholarship. If they choose to skip a scholarship k , they lose its importance rating w_k , but they no longer need to worry about its open interval intersecting their chosen period.

The total cost of a schedule is defined as the sum of:

- The length of the chosen period $[L, R]$, calculated as $R - L + 1$.
- The importance rating w_k of the skipped scholarship (if one was skipped; otherwise 0).

Your goal is to determine the minimum possible total cost the student can achieve.

Constraints

- $1 \leq N \leq 10^6$
- $1 \leq l_i \leq r_i \leq 10^9$
- $1 \leq w_i \leq 10^9$

Input

The first line of the input contains a single integer N , the number of scholarships. The next N lines each contain three space-separated integers l_i , r_i , and w_i , representing the start day, end day, and the importance rating of the i -th scholarship.

Output

The only line of output contains the minimum cost achievable.

Subtasks

Test group	Points	Constraints
1	10	$N \leq 8, r_i \leq 20$
2	15	$N \leq 2000, r_i \leq 2000$, and all $w_i = 2000$
3	15	$N \leq 2000, r_i \leq 2000$
4	15	$N \leq 2 \cdot 10^5, r_i \leq 10^9$, and all $w_i = 10^9$
5	20	$N \leq 2 \cdot 10^5, r_i \leq 10^9$
6	25	$N \leq 5 \cdot 10^6, r_i \leq 10^9$

Examples

Example 1

Input

```
3
1 5 10
7 10 10
4 8 2
```

Output

```
3
```

Explanation

It is optimal to skip no scholarships. The minimum length period intersecting all three intervals is $[5, 7]$, which has a length of $7 - 5 + 1 = 3$. The cost is $3 + 0 = 3$.

Example 2

Input

```
3
4 6 10
4 6 10
100 100 2
```

Output

```
3
```

Explanation

It is optimal to skip the third scholarship, losing its rating of 2. The remaining intervals are $[4, 6]$ and $[4, 6]$. We can pick the period $[5, 5]$ which intersects both and has a length of 1. The cost is $1 + 2 = 3$.

Example 3

Input

```
3
1 5 10
3 8 10
4 10 10
```

Output

```
1
```

Explanation

All three intervals overlap at day 4 and day 5. We can pick a 1-day period like $[4, 4]$ intersecting all of them without skipping anything. The cost is $1 + 0 = 1$.